

Design a URL Shortener - Step 1: Understand the Problem & Scope

1. Problem Overview

- Design a URL shortening service like TinyURL

2. Example

None

Original URL:

<https://www.systeminterview.com/q=chatsystem&c=loggedin&v=v3&l=long>

Short URL:

<https://tinyurl.com/y7keocwj>

Clicking short URL → **redirects to original URL**

3. Clarification Questions

Traffic Volume

- 100 million URLs/day

URL Length

- Should be **as short as possible**

Allowed Characters

None

0-9, a-z, A-Z

Data Modification

- Shortened URLs:
 - Cannot be deleted
 - Cannot be updated

4. Core Use Cases

1. URL Shortening

None

Long URL → Short URL

2. URL Redirecting

None

Short URL → Original URL

3. System Requirements

- High availability
- Scalability
- Fault tolerance

5. Key Takeaway

System must efficiently handle:

- Massive scale (100M/day)
- Fast redirection
- Reliable storage of mappings